# Ticket 2

TCK02-AccLock

## Summary

| | |
|---|---|
| Category | Account management |
| Requester | Carla Reyes (Sales) |
| Request | "I can't sign in, my password isn't working" |
| Action | Confirm the lockout, trace the source, unlock the account, verify |
| Tools | Active Directory Users and Computers (ADUC), PowerShell, Event Viewer logs |
| Outcome | Resolved. Account unlocked and user signed in successfully |

## Problem

Carla Reyes (Sales) reported "I can't sign in, my password isn't working". At sign-in she sees the message "The referenced account is currently locked out and may not be logged on to" (Figure 1). Before changing anything, the requester's identity was verified with their employee ID.

**Background: what a lockout is.** Active Directory counts failed password attempts on every account. When the count reaches the limit, the account is **locked**, and AD rejects **even the correct password** until the account is unlocked or the timer expires. It exists to stop brute-force password guessing.

## Environment

| Policy setting | Value |
|---|---|
| Lockout threshold | 5 failed attempts |
| Lockout duration | 30 minutes (then it unlocks itself) |
| Observation window | 30 minutes (the failed-attempt counter resets after this) |

| Item | Detail |
|---|---|
| Domain | acme.local |
| Account | creyes (Carla Reyes), Sales |
| Client | DESKTOP-HEOKDBL, joined to acme.local |
| Domain controller | WIN-4G8SQK9BMC1 |
| Tools | ADUC, PowerShell, Event Viewer logs |

## Troubleshooting Steps

**Triage: is it really a lockout?** "My password isn't working" has several causes with different fixes. The message at sign-in tells them apart:

| Message at sign-in | Cause | Fix |
|---|---|---|
| "The referenced account is currently locked out and may not be logged on to" | **Locked out** | Unlock (this ticket) |
| "The user name or password is incorrect" | Wrong password, not locked | Retry carefully, or reset (Ticket 01) |
| "Your account has been disabled" | Account disabled (e.g. a leaver) | Enable, with approval |

| Message at sign-in | Cause | Fix |
|---|---|---|
| "Your password has expired and must be changed" | Expired password | User changes it at the prompt |

| # | Action | Result |
|---|---|---|
| 1 | Verify the requester with their employee ID. | Identity verified. |
| 2 | Client: read the sign-in message. | Referenced account is currently locked out. |
| 3 | Server: `Get-ADUser creyes -Properties LockedOut,BadLogonCount,LastBadPasswordAttempt \| Select-Object Name,LockedOut,BadLogonCount,LastBadPasswordAttempt`. The message alone isn't proof, so check the account. | `LockedOut` is `True` and `BadLogonCount` is 5, which matches the threshold. |
| 4 | Find the source of the failed attempts. Unlocking without finding the cause can lead to the account locking again. Event **4740** is logged on the domain controller each time an account locks: `Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4740} -MaxEvents 5 \| Format-List TimeCreated,Message` | Event 4740 shows the locked account and the caller computer name. |
| 5 | Confirm the caller is the user's own machine: System properties on the client. | Computer name DESKTOP-HEOKDBL on acme.local matches the caller. The failed attempts came from the user's own PC, not an unknown device, so this isn't a sign of an attack. |
| 6 | Unlock the account (see Fix). | Account unlocked. |
| 7 | Verify in AD and on the client (see Verification). | Resolved. |

**How to read event 4740:**

| Field | Meaning |
|---|---|
| **Subject → Account Name** (`WIN-4G8SQK9BMC1$`) | The domain controller that recorded the event. Not the culprit. |
| **Account That Was Locked Out → Account Name** (`creyes`) | The affected user |
| **Caller Computer Name** (`DESKTOP-HEOKDBL`) | The machine that sent the failed attempts |
| **TimeCreated** (`7:56:21 PM`) | Matches `LastBadPasswordAttempt` from step 3 |

## Root Cause

In the lab the failed attempts were sent from the client PC. In a real environment the usual causes are:

| Cause | How to recognise it | Fix |
|---|---|---|
| Typo, caps lock, wrong keyboard layout | One-off lockout | Unlock; user retries carefully |
| **Stale saved password** on a phone, second PC, mapped drive, VPN, or saved Wi-Fi after a password change | **Locks again minutes after unlocking** | Update the password on that device |
| Service or scheduled task running as the user with an old password | Locks on a regular timer | Update the stored password, or move the job to a service account |
| Someone guessing passwords | Many accounts locking at once, or lockouts at odd hours | **Escalate to security.** Don't just unlock. |

When the caller in event 4740 points to a server (a mail server or a RADIUS server, say), that server is a middleman. Check its logs to find the device behind it.

## Fix

In ADUC: right-click the user → **Properties** → **Account** tab → tick **Unlock account** → OK. The checkbox only appears while the account is locked, which is another confirmation.

PowerShell equivalent:

```
Unlock-ADAccount -Identity creyes
```

After a password reset, tell the user to update the password on every device they use. That prevents most repeat lockouts.

## Verification

**In AD:** the same command now shows `LockedOut` as `False` and `BadLogonCount` back at `0`. `LastBadPasswordAttempt` keeps its old timestamp because it records history.

**On the client:** the user signs in and the session starts. `whoami` in the new session confirms the signed-in identity is the domain account (`acme\creyes`).

## Lessons Learned

• **Unlocking doesn't fix the cause.** If the account re-locks, something is still sending the old password.

• **The Reset Password dialog has an unlock checkbox.** If the user locked themselves out while trying to remember a forgotten password, one action can reset and unlock.

• **A reset is not an unlock, and an unlock is not a reset.** They are separate actions.

• With multiple domain controllers, bad-attempt counters are tracked per DC and can look inconsistent. Query the PDC emulator for the authoritative count.

**Command reference**

```
# Check one user
Get-ADUser <user> -Properties LockedOut,BadLogonCount,LastBadPasswordAttempt |
  Select-Object Name,LockedOut,BadLogonCount,LastBadPasswordAttempt

# List every locked account in the domain
Search-ADAccount -LockedOut | Select-Object Name,SamAccountName

# Unlock
Unlock-ADAccount -Identity <user>

# Find the source of the lockout (run on the DC, elevated)
Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4740} -MaxEvents 5 |
  Format-List TimeCreated,Message
```

## Screenshots

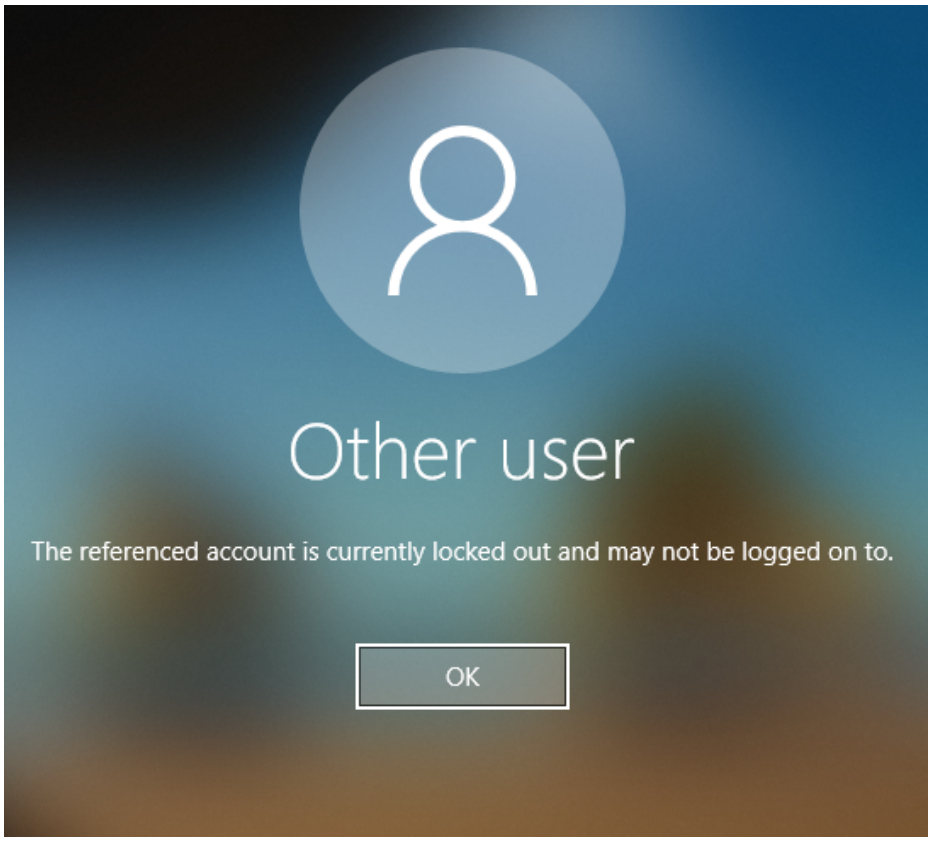



*Figure 1. Sign-in error: the referenced account is currently locked out*

```
PS C:\Users\Administrator>  Get-ADUser creyes -Properties LockedOut,BadLogonCount,LastBadPasswordAttempt |
 Select-Object Name,LockedOut,BadLogonCount,LastBadPasswordAttempt

Name        LockedOut BadLogonCount LastBadPasswordAttempt
----        --------- ------------- ----------------------
Carla Reyes      True             5 10/1/2026 7:56:21 PM
```

*Figure 2. Get-ADUser output before the unlock: LockedOut True, BadLogonCount 5*

```
PS C:\Users\Administrator> Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4740} -MaxEvents 5
| Format-List TimeCreated,Message


TimeCreated : 10/1/2026 7:56:21 PM
Message     : A user account was locked out.

              Subject:
                Security ID:            S-1-5-18
                Account Name:           WIN-4G8SQK9BMC1$
                Account Domain:         ACME
                Logon ID:               0x3E7

              Account That Was Locked Out:
                Security ID:            S-1-5-21-242057048-1441979557-2913986384-1110
                Account Name:           creyes

              Additional Information:
                Caller Computer Name:   DESKTOP-HEOKDBL
```

*Figure 3. Event 4740 showing the locked account and the caller computer name*

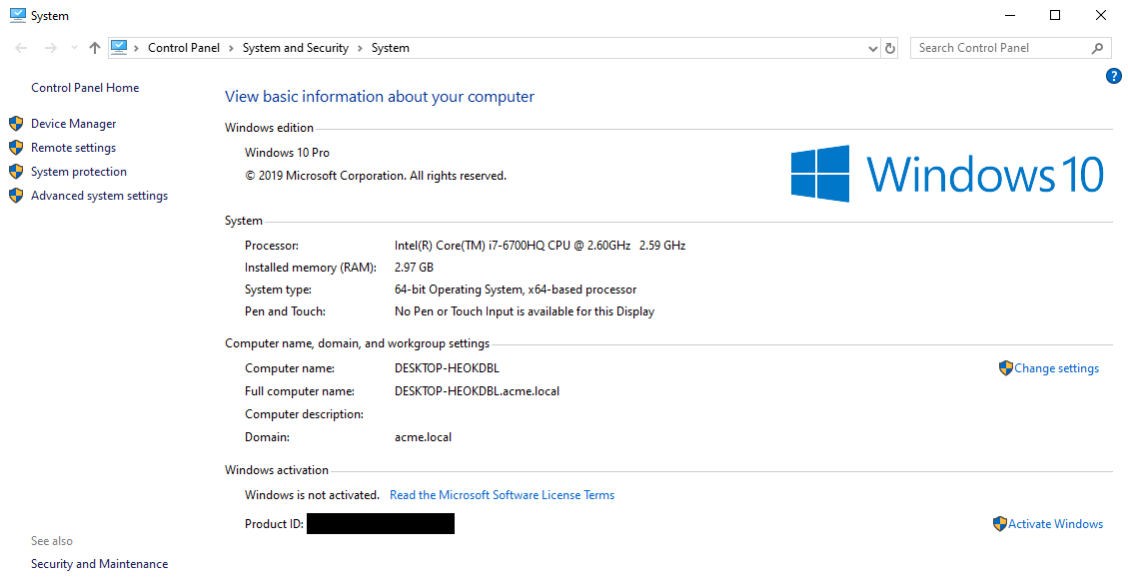



*Figure 4. System properties showing computer name DESKTOP-HEOKDBL on the acme.local domain*

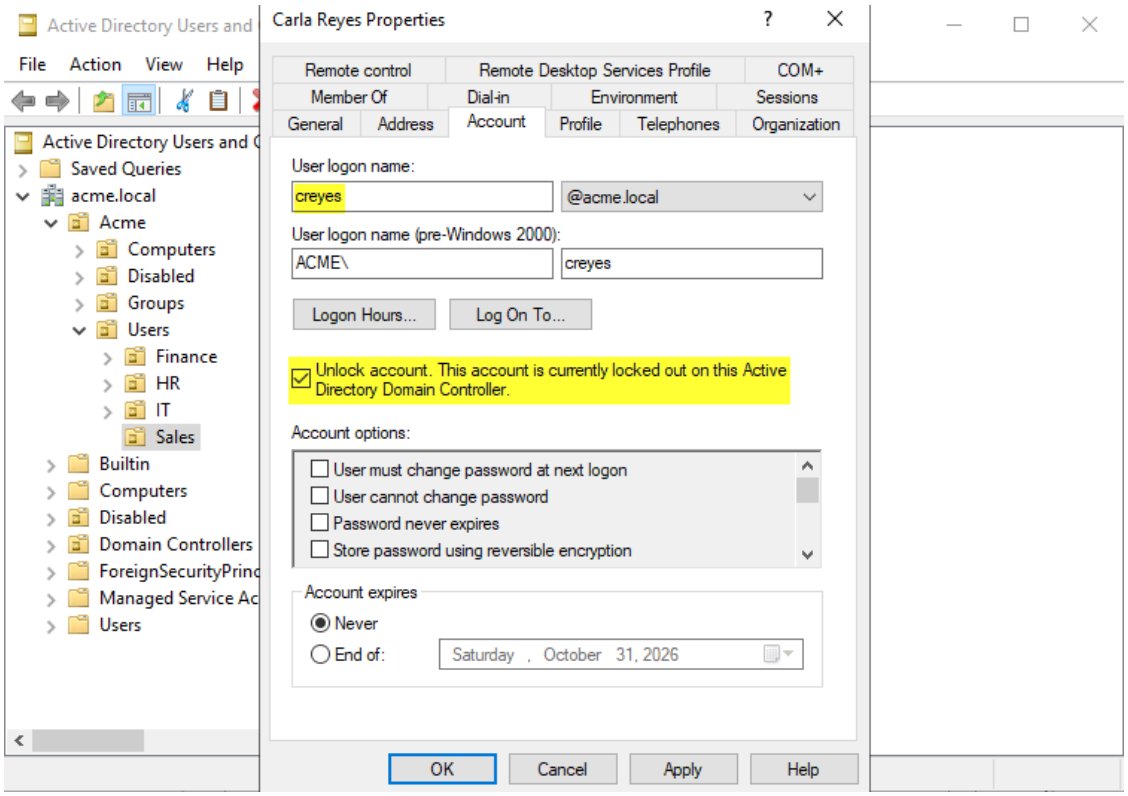



*Figure 5. ADUC Account tab with the Unlock account checkbox ticked*

```
PS C:\Users\Administrator>  Get-ADUser creyes -Properties LockedOut,BadLogonCount,LastBadPasswordAttempt |
 Select-Object Name,LockedOut,BadLogonCount,LastBadPasswordAttempt

Name         LockedOut BadLogonCount LastBadPasswordAttempt
----         --------- ------------ ----------------------
Carla Reyes     False             0 10/1/2026 7:56:21 PM
```

*Figure 6. Get-ADUser output after the unlock: LockedOut False, BadLogonCount 0*

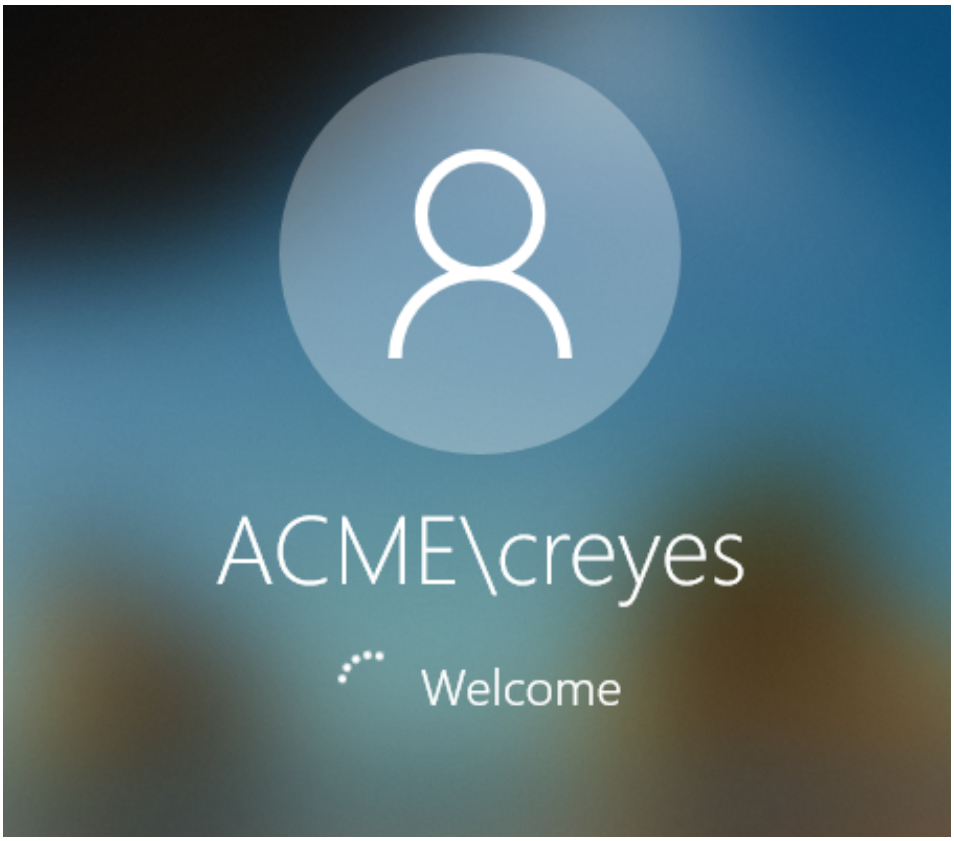



*Figure 7. Welcome screen after a successful sign-in as ACME\creyes*

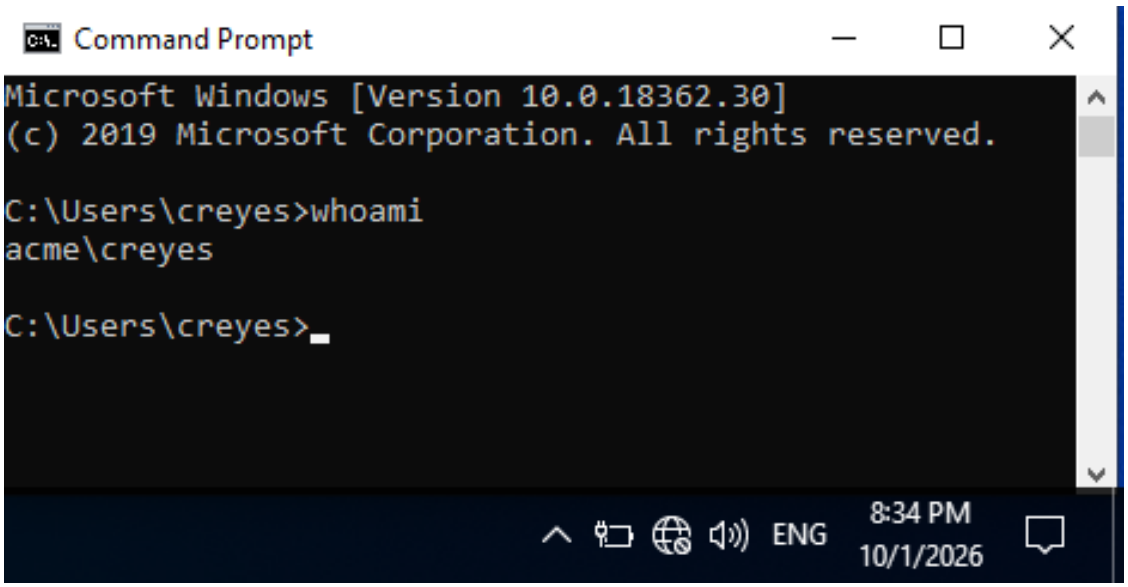



*Figure 8. Command prompt: whoami returns acme\creyes*